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Editors

Journal of Open Research Software

Dear Editors,

Please consider the enclosed software metapaper, "Hephaestus: an in-browser open-source engine and databases for the thermodynamics and phase diagrams of slags, molten salts, and alloys," for publication in the Journal of Open Research Software.

The submission describes two coupled research artifacts. The first is a dependency-free C implementation of the MQMQA thermodynamic model with a Python interface and a WebAssembly build, so the same validated code runs embedded, scripted, or in a browser with nothing installed. The second is, to the authors' knowledge, the first freely licensed MQMQA oxide slag database, assembled entirely from published measurements with per-value provenance. The engine reads the field's four database containers, ChemSage .dat, Thermo-Calc .tdb with the Inden magnetic model, the openly specified uTDB unified dialect, and the XML XTDB successor with its third-generation models, demonstrated by shipped aluminum-recycling and steelmaking files that each carry an alloy together with its salt or slag; an open chloride molten-salt family ships alongside the slags. Every energy path is validated to machine precision against pycalphad, which was itself published in this journal, and the assembled ternary database is tested against the classical measured phase diagram of the MgO-FeO-SiO2 system; the magnetic model is additionally checked against the measured pure-iron transition temperatures.

The software is MIT licensed, the database is CC BY 4.0, and both are in active use. The manuscript follows the journal's metapaper structure, and the version described is archived as release v0.6.0 of the public repository.

This work is original, is not under consideration elsewhere, and declares its competing interest in the manuscript. Thank you for your consideration.

Sincerely,

Michael E. Bustamante

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